Hybrid Warehouse Delivery Agent

Mini-Project Report

This report covers the warehouse delivery mini-project only.

1. Problem and objective

A warehouse robot collects items from shelves A, B, and C and returns each item to the dock while human workers move unpredictably. The goal is to complete ten orders with fewer steps and waits, appropriate re-planning, and no worker-cell entries in the simulation.

2. PEAS

Component	Description
Performance	Complete orders; zero simulated collisions; reduce steps, waits, and re-plans.
Environment	Grid warehouse, shelves A/B/C, dock, moving workers, static obstacles.
Actuators	Move Up, Down, Left, Right; wait.
Sensors	Robot location, known grid/shelf/dock locations, current worker positions.

3. Environment classification: seven dimensions

1. Partially observable: current worker positions are observed, but future positions are unknown.

2. Multi-agent: the robot shares the space with human workers.

3. Stochastic: worker movement has random choices.

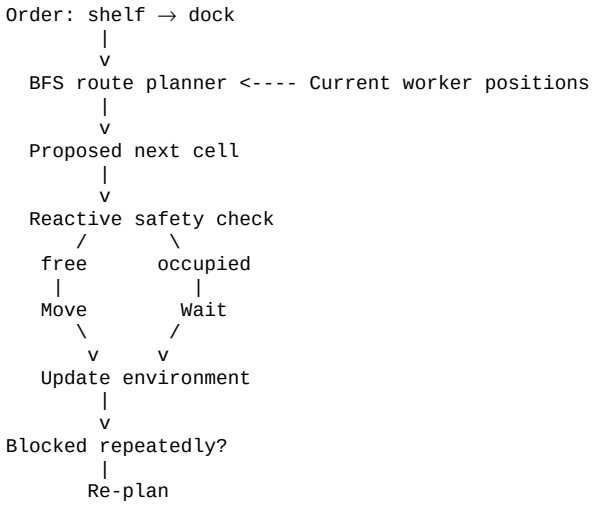
4. Sequential: each move affects later states and decisions.

5. Dynamic: workers move during operation.

6. Discrete: grid cells and actions are discrete.

7. Known layout and robot transitions, uncertain future worker behavior.

4. Architecture



BFS finds a shortest path on the unweighted grid. The reactive layer checks the next cell before movement; if occupied, the robot waits. After two consecutive blocked attempts, it re-plans using current worker positions.

5. Evaluation

Seed	Steps	Waits	Re-plans	Collisions	Delivered
0	223	27	32	0	10
1	183	11	25	0	10
2	227	27	32	0	10

Ten shelf orders were run for each of three fixed seeds. Steps include movement and waiting steps. Results are reproducible for this implementation and these seeds.

6. Discussion

A reactive-only agent may avoid an immediate obstacle but navigate inefficiently. A fixed plan may become stale as workers move. The hybrid combines shortest-path planning with an immediate safety check and re-plans after repeated blocking.

7. Limitations and safety

The simulation assumes accurate sensing of current worker positions and simplified random movement. It does not model robot size, braking distance, occlusion, sensor delay, or realistic human motion. Zero simulated collisions is not a real-world safety guarantee. Physical deployment would require robust sensing, safety-rated stopping, conservative distance margins, extensive testing, and human oversight.